

KNOWLEDGE INFERENCE LAB REPORT

Course Code: APT 3020

Assignment: Lab 2: Reasoning and Inferencing in Expert Systems

Project Title: Building a Reasoning and Inferencing Engine for a Student Academic Advisor

Executive Project Summary

This project presents the implementation of a rule-based reasoning engine designed to evaluate university student profiles. The system collects student input data and transforms it into structured facts. These facts are then processed using a forward chaining inference mechanism to generate conclusions based on predefined rules in a knowledge base.

Knowledge Representation

The system represents knowledge using a JSON-based rule structure. Each rule consists of:

- Conditions (facts that must be satisfied)
- A conclusion (result if conditions are met)

Facts are derived from user input such as GPA, attendance, disciplinary status, prerequisite completion, and fee status.

Reasoning Method (Forward Chaining)

The system uses a **data-driven forward chaining approach**, which works as follows:

1. Accept user input
2. Convert input into structured facts
3. Compare facts with rule conditions
4. Trigger rules whose conditions are satisfied
5. Generate conclusions

Forward Chaining Inferencing Demonstration

Test Case

- Facts provide (Input):

- GPA: 3.6
- Attendance: 90%
- Disciplinary Case: No
- Completed Prerequisite: Yes
- Outstanding Fees: No
- Rules triggered:
 - Scholarship rule
 - Graduation rule
 - Dean's list rule
- Conclusions made:
 - Eligible for Scholarship because:
 - ✓ GPA > 3.5
 - ✓ Attendance > 80%
 - ✓ No disciplinary cases
 - Eligible for Graduation because:
 - ✓ GPA > 3.0
 - ✓ Completed Prerequisite Courses
 - ✓ No Outstanding Fees
 - Dean's List Candidate because:
 - ✓ GPA > 3.5
 - ✓ Attendance > 80%

The system includes an explanation feature that shows:

- Which rules were triggered
- Which conditions were satisfied

This improves transparency and allows users to understand how conclusions were reached.

Challenges Faced

- Ensuring consistency between JSON rules and Python facts
- Handling input transformation correctly
- Debugging missing or mismatched keys

Conclusion

The system successfully demonstrates rule-based reasoning using forward chaining. It provides clear conclusions along with explanations, making it useful for decision-support applications.